



COMPETENCY STANDARD

FOR

**IT SUPPORT TECHNIQUE & MOBILE
PHONE SERVICING**

(ICT SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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The Competency Standards for IT Support Technique & Mobile Phone Servicing serve as a foundational document for the development of curricula, teaching materials, and assessment tools specifically tailored for the ICT sector. This document ensures that training activities align with industry requirements, enabling individuals who successfully meet the established standards through assessment to become qualified professionals in relevant roles within the sector.

Developed under the Skills for Industry Competitiveness and Innovation Program (SICIP), this document addresses the skills requirements critical for IT Support Technique & Mobile Phone Servicing. It is owned by the Finance Division of the Ministry of Finance of the People's Republic of Bangladesh.

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INTRODUCTION

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The Competency Standard also suggests integration of YouTube or similar digital platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution and relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Identification and validation of units of competency and elements for each occupation were made by experts from various companies in industry consultative workshops.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:**Generic Competencies (20 hrs.)**

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-ICT-ITM-01-G	Apply Occupational Health and Safety (OHS) Practices in the Workplace	1. Identify OHS policies and procedures 2. Apply personal health and safety procedures 3. Report hazards and risks 4. Respond to emergencies	10
SICIP-ICT-ITM-02-G	Carryout Workplace Interaction in English	1. Interpret workplace communication and etiquette 2. Read and understand workplace documents 3. Participate in workplace meetings and discussions 4. Practice professional ethics at workplace	10
Total Hour			20 Hrs.

Sector Specific Competencies (30 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-ICT-ITM-01-S	Operate Office Application Software	1. Operate computer 2. Install application software 3. Use word processor to prepare/create documents 4. Use spreadsheet to create/prepare worksheets 5. Use presentation software to create/prepare presentation	20
SICIP-ICT-ITM-02-S	Comply with Ethical Standards in IT Workplace	1. Uphold the requirements of clients 2. Deliver quality products and services 3. Maintain professionalism at workplace 4. Maintain workplace code of conduct	10
Total Hour			30 Hrs.

Occupation Specific Competencies (310 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-ICT-ITM-01-O	Interpret IT and Electronics Fundamentals	1. Interpret basic electrical and electronics concepts. 2. Identify computer components. 3. Identify mobile phone components. 4. Interpret mobile phone circuit and schematic diagrams.	20
SICIP-ICT-ITM-02-O	Assemble and Configure Computer Systems	1. Prepare tools, workspace, and components. 2. Disassemble and assemble computer hardware. 3. Configure BIOS and system settings. 4. Install operating systems, drivers, and software. 5. Optimize system performance and updates.	40
SICIP-ICT-ITM-03-O	Manage Network and Internet Connectivity	1. Set up local area network (LAN) and Wi-Fi connections. 2. Configure network devices. 3. Monitor network performance. 4. Apply troubleshooting methods for network failures. 5. Secure network access and antivirus tools.	40
SICIP-ICT-ITM-04-O	Maintain Data Backup and System Security	1. Identify critical data and establish backup schedules. 2. Perform data backup using appropriate storage methods (cloud or local). 3. Test backup integrity to ensure data. 4. Recover lost or corrupted data. 5. Implement disaster recovery procedures	60
SICIP-ICT-ITM-05-O	Practice Soldering and	1. Select soldering tools and materials 2. Perform soldering on PCB	40

	De-soldering	3. Perform de-soldering components from PCB 4. Perform re-soldering from PCB 5. Clean soldering tools and workspace	
SICIP-ICT-ITM-06-O	Test components and service mobile phone	1. Prepare mobile phone, tools, and workspace for servicing. 2. Apply standard testing methods for components. 3. Disassemble the mobile phone. 4. Test components in-circuit and off-circuit. 5. Replace faulty or damaged components. 6. Reassemble the mobile phone.	70
SICIP-ICT-ITM-07-O	Apply Software Installation and Flashing Mobile Phone	1. Identify software requirements of mobile phone. 2. Prepare flashing tools and install drivers. 3. Connect mobile phone to PC or flashing tool. 4. Flash mobile phone software. 5. Clean tools and equipment.	40
Total Hour			310 Hrs.

COMPETENCY STANDARD: IT SUPPORT TECHNIQUE & MOBILE PHONE SERVICING

The Generic Competencies

Unit of Competency: APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICES IN THE WORKPLACE	Nominal Duration: 10 hrs.	Unit Code: SICIP-ICT-ITM-01-G
Unit Descriptor: This unit covers knowledge, skills and attitudes required to apply occupational health and safety (OHS) practices in the workplace. It specifically includes the tasks of identifying OHS policies and procedures, applying personal health and safety procedures, reporting hazards and risks, and responding to emergencies.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1.1 <u>OHS policies</u> and safe operating procedures are read and understood. 1.2 Safety signs and symbols are identified and followed. 1.3 Emergency response, evacuation procedures and other contingency measures are determined.
2. Apply personal health and safety procedures	2.1 OHS policies and procedures are practiced and applied. 2.2 <u>Personal Protective Equipment (PPE)</u> is selected and used. 2.3 Personal hygiene is practiced.
3. Report hazards and risks	3.1 <u>Hazards and risks</u> are identified, assessed and controlled. 3.2 Incidents arising from hazards and risks are reported to authority. 3.3 Corrective actions are implemented to correct unsafe conditions in the workplace.
4. Respond to emergencies	4.1 Alarms and warning devices are responded. 4.2 <u>Emergency response plans and procedures</u> are implemented. 4.3 First aid procedure is applied during emergency situations.

Range of Variables

Variable	Range
	May include but not limited to:
1. OHS policies	1.1 International/ Local OHS requirements 1.2 Bangladesh Standards for OHS Building Code 1.3 Fire Safety Rules and Regulations
2. Personal Protective	2.1 Apron

Equipment (PPE)	2.2 Gas Mask 2.3 Gloves 2.4 Safety shoes 2.5 Face mask 2.6 Goggles 2.7 Ear plugs 2.8 Scarf
3. Hazards and risks	3.1 Chemical hazards 3.2 Biological hazards 3.3 Physical Hazards 3.4 Machine hazards 3.5 Materials hazards 3.6 Tools and Equipment hazards
4. Emergency response plans and procedures	4.1 Firefighting procedures 4.2 Earthquake response procedures 4.3 Evacuation procedures 4.4 Medical and first-aid

Curricular Content Guide:

1. Underpinning Knowledge	1.1 OHS workplace policies and procedures 1.2 Work safety procedures 1.3 Emergency procedures – firefighting, earthquake response, explosion response & accident response 1.4 Types of hazards – biological, chemical & physical, and their effects 1.5 PPE types and uses 1.6 Personal hygiene practices 1.7 OHS awareness 1.8 Malfunctions and resolutions
2. Underpinning Skills	2.1 Identifying OHS policies and procedures 2.2 Applying personal work safety practices in the workplace 2.3 Identifying, assessing and controlling hazards and risks 2.4 Reporting incidents arising from hazards and risks 2.5 Responding to emergency procedures 2.6 Maintaining physical well-being in the workplace 2.7 Implementing emergency response plans and procedures 2.8 Applying basic first aid procedures
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace

4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 PPEs 4.3 Firefighting equipment 4.4 Emergency response manual 4.5 First aid kits 4.6 Learning manual
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Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 applied OHS policies and procedures 1.2 selected and used personal protective equipment (PPE) 1.3 practiced personal health and safety procedures 1.4 identified, assessed and controlled hazards and risks 1.5 reported incidents arising from hazards and risks to the authority 1.6 implemented plans and procedures to respond emergencies 1.7 applied basic first aid procedure
2. Methods of Assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral question 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert.

Unit of Competency: CARRY OUT WORKPLACE INTERACTION IN ENGLISH	Nominal Duration: 10 hrs.	Unit Code: SICIP-ICT-ITM-02-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to carry out workplace interaction in English. It specifically includes the tasks of interpreting workplace communication and etiquette, reading and understanding workplace documents, participating in workplace meetings and discussions, and practicing professional ethics at workplace.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret workplace communication and etiquette	1.1 Workplace code of conducts are interpreted as per organizational guidelines. 1.2 Appropriate lines of communication are maintained with supervisors and colleagues. 1.3 Workplace interactions are conducted in a <u>courteous manner</u> to gather and convey information. 1.4 Questions about routine <u>workplace procedures and matters</u> are asked and responded as required.
2. Read and understand workplace documents	2.1 Workplace documents are interpreted as per standard. 2.2 Assistance is taken to aid comprehension when required from peers/supervisors. 2.3 Visual information/symbols/signages are understood and followed. 2.4 Specific and relevant information are accessed from <u>appropriate sources</u> . 2.5 Appropriate medium is used to transfer information and ideas.
3. Participate in workplace meetings and discussions	3.1 Team meetings are attended on time and followed meeting procedures and etiquette. 3.2 Own opinions are expressed and listened to those of others without interruption. 3.3 Inputs are provided consistent with the meeting purpose and interpreted and implemented meeting outcomes.
4. Practice professional ethics at workplace	4.1 Responsibilities as a team member are demonstrated and kept promises and commitments made to others. 4.2 Tasks are performed in accordance with workplace procedures. 4.3 Confidentiality is respected and maintained. 4.4 Situations and actions considered inappropriate or which present a conflict of interest are avoided.

Range of Variables

Variable	Range May Include but not limited to:
1. Courteous manner	1.1 Effective questioning 1.2 Active listening 1.3 Speaking skills
2. Workplace procedures and matters	2.1 Notes 2.2 Agenda 2.3 Simple reports such as progress and incident reports 2.4 Job sheets 2.5 Operational manuals 2.6 Brochures and promotional material 2.7 Visual and graphic materials 2.8 Standards 2.9 OSH information 2.10 Signs
3. Appropriate sources	3.1 HR Department 3.2 Managers 3.3 Supervisors

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Workplace code of conducts 1.2 Workplace documents, signs and symbols 1.3 Meeting procedure and etiquette
2. Underpinning Skills	2.1 Demonstrating performance of workplace communication and etiquette 2.2 Applying workplace information and symbols/signages 2.3 Applying workplace code of conducts as per organizational guidelines 2.4 Providing inputs in the meeting consistent with the meeting purpose 2.5 Interpreting and implementing meeting outcomes 2.6 Practicing professional ethics at workplace 2.7 Avoiding tactfully situations and actions considered inappropriate or which presents a conflict of interest
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided:

	4.1 Workplace (simulated or actual) 4.2 Work book 4.3 Learning & operational manuals 4.4 Workplace tools/equipment 4.5 Relevant tools, equipment, software and facilities needed to perform the activities
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Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted workplace communication and etiquette 1.2 practiced workplace code of conducts as per organizational guidelines 1.3 Interpreted workplace documents, instructions and symbols 1.4 participated in workplace meetings and discussions 1.5 interpreted and implemented meeting outcomes 1.6 practiced professional ethics at workplace
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert

The Sector Specific Competencies

Unit of Competency: OPERATE OFFICE APPLICATION SOFTWARE	Nominal Duration: 20 hrs.	Unit Code: SICIP-ICT-ITM-01-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to operate office application software. It specifically includes the tasks of operating computer, installing application software, using word processor to prepare/create documents, using spreadsheet to create/prepare worksheets and using presentation software to create/prepare presentation.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Operate computer	1.1 Safe workplace practices are observed according to IT workplace guideline. 1.2 Desktop <u>Peripherals</u> are checked and connected with computer properly. 1.3 Computer is switched on. 1.4 Computer <u>desktop/GUI settings</u> are arranged and customized as per requirement. 1.5 Files and folders are <u>manipulated</u> as per requirement. 1.6 Properties of files and folders are viewed and searched. 1.7 Disks are defragmented, formatted as per requirement.
2. Install application software	2.1 Installation requirements of software are identified and listed. 2.2 Software sources and CD key/ password are assured. 2.3 <u>Appropriate Software</u> are collected and selected as per requirement. 2.4 Software installation is started. 2.5 Customization is done as per requirement. 2.6 Steps of installation are followed as per installation Instructions. 2.7 Installations are completed properly. 2.8 Correctness of Installation is checked.
3. Use word processor to prepare/create documents	3.1 Appropriate <u>word processor</u> is selected and started. 3.2 Documents are created as per requirement in personal use and office environment. 3.3 Contents are entered. 3.4 Documents are formatted. 3.5 Paragraph and page settings are completed. 3.6 Document is saved.
4. Use spreadsheet to create/prepare	4.1 <u>Spreadsheet applications</u> are selected and started. 4.2 Worksheets are created as per requirement in personal

worksheets	use and office environment. 4.3 Data are entered. 4.4 Functions are used for calculating and editing logical operation. 4.5 Sheets are formatted as per requirement. 4.6 Charts are created. 4.7 Charts/sheets are saved.
5. Use presentation software to create/prepare presentation	5.1 Appropriate <u>presentation applications</u> are selected and started. 5.2 Presentations are created as per requirement in personal use and office environment 5.3 Image, illustrations, text, table, symbols and media are entered as per requirements. 5.4 Presentations are formatted and animated. 5.5 Presentations are viewed and saved.

Range of Variables

Variable	Range
	May include but not limited to:
1. Peripherals	1.1 Monitor 1.2 Keyboard 1.3 Mouse 1.4 Modem 1.5 Scanner 1.6 Printer
2. Desktop settings	2.1 Icons 2.2 Taskbar 2.3 View 2.4 Resolutions
3. Manipulate	3.1 Create 3.2 Open 3.3 Copy 3.4 Rename 3.5 Delete 3.6 Sort
4. Typing tutors	4.1 English typing tutor 4.2 Bangla typing tutor
5. Appropriate software	5.1 MS office or Open office but limited to 5.2 Word processor software. 5.3 Spread sheet software 5.4 Presentation software
6. Word processor	6.1 MS Word processor 6.2 Open office Org 6.3 Google docs 6.4 Word perfect 6.5 LibreOffice
7. Spread sheet	7.1 MS Excel

applications	7.2 Google Sheets 7.3 Apple Numbers by Apple
8. Presentation application	8.1 MS PowerPoint 8.2 Google Slides 8.3 Prezi

Curricular Content Guide

1. Underpinning Knowledge	1.1 Desktop items 1.2 Type of Bangla keyboard layout 1.3 Different types of software and application packages 1.4 Use of word processor, spread sheet and presentation software 1.5 Types of printers 1.6 Types of charts, impotence of chart 1.7 Different types of math and logical functions
2. Underpinning Skills	2.1 Starting computer 2.2 Installing operating system 2.3 Managing desktop item 2.4 Manipulating files and folders as per requirement 2.5 Installing application software 2.6 Running application software 2.7 Creating and saving document with word processing application 2.8 Using functions in spread sheet. 2.9 Applying animations into presentation slide 2.10 Printing document
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Computer/laptop 4.3 Scanner, printer 4.4 Software 4.5 Work books 4.6 Relevant tools, equipment and facilities needed to perform the activities 4.7 Required learning materials

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate:
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	1.1 installed operating system 1.2 manipulated files and folders as per requirement 1.3 Installed appropriate application software 1.4 used word processor to prepare/create documents 1.5 used spreadsheet to create/prepare worksheets 1.6 Used presentation software to create/prepare presentation 1.7 applied animations into presentation slides 1.8 printed document
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: COMPLY WITH ETHICAL STANDARDS IN IT WORKPLACE	Nominal Duration: 10 hrs.	Unit Code: SICIP-ICT-ITM-02-S
Unit Descriptor: This unit covers knowledge, skills and attitudes required to comply with ethical standards in IT workplace. It specifically includes the tasks of upholding the requirements of clients, delivering quality products and services, maintaining professionalism at workplace and maintaining workplace code of conduct.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined>** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Uphold the requirements of clients	1.1 Clients' requirements are identified. 1.2 Confidentiality of information is maintained in accordance with workplace policies/organizational policies/national legislation. 1.3 Potential conflicts of interest are identified and involved parties of potential conflicts are notified. 1.4 Proprietary rights of client/customer are asserted.
2. Deliver quality products and services	2.1. Products and services are provided according to the clients' requirements. 2.2. Work is completed as per standards. 2.3. Quality processes are implemented when developing products and services.
3. Maintain professionalism at workplace	3.1 Work processes are delivered as per standards. 3.2 Skills, knowledge and qualifications are presented in a professional manner. 3.3 Services and products developed by self and others are delivered as per workplace standard. 3.4 Unbiased and objective information are provided to the clients. 3.5 Realistic estimates for time, cost and delivery of outputs are presented during negotiation.
4. Maintain workplace code of conduct	4.1 Workplace code of conduct is interpreted. 4.2 Workplace code of conduct is followed.

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Corporate code of confidentiality of information 1.2 Organizational policies, national legislation and workplace policies in relation to IT sector 1.3 Law and regulations pertaining to proprietary rights 1.4 Quality processes for products and services 1.5 Procedure of provided to client information 1.6 Method of estimating for time, cost and delivery products
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	and services 1.7 Workplace code of conduct in IT sector
2. Underpinning Skills	2.1 Upholding confidentiality of information in accordance with organizational policies, national legislation and workplace policies 2.2 Asserting proprietary rights of clients/customers 2.3 Completing work in accordance with industry and international standards 2.4 Implementing quality processes when developing products and services 2.5 Delivering correctly services and products developed by self and others 2.6 Providing unbiased and objective information to the clients 2.7 Presenting realistic estimates for time, cost and delivery of outputs during negotiation 2.8 Maintaining workplace code of conduct
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Computer/laptop 4.3 National legislation in relation to IT sector 4.4 Workplace policies, code of conduct & proprietary rights of clients in IT sector 4.5 Relevant tools, equipment and facilities needed to perform the activities 4.6 Required learning materials

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 upheld the requirements of clients/customers 1.2 asserted proprietary rights of clients/customers 1.3 delivered quality products and services 1.4 Implemented quality processes when developing products and services 1.5 maintained professionalism at the workplace 1.6 completed work according to industry and international standards 1.7 maintained workplace code of conduct
2. Methods of	Methods of assessment should include:

Assessment	2.5 Written test 2.6 Practical demonstration 2.7 Oral question 2.8 Portfolio (optional)
3. Context of Assessment	3.3 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training 3.4 Assessment should be done by a certified assessor or occupation-specific industry expert.

The Occupation Specific Competencies

Unit of Competency: INTERPRET IT AND ELECTRONICS FUNDAMENTALS	Nominal Duration: 20 hrs.	Unit Code: SICIP-ICT-ITM-01-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to interpret IT and electronics fundamentals. It specifically includes the tasks of interpreting basic electrical and electronics concepts, identifying computer components, identifying mobile phone components and interpreting mobile phone circuit and schematic diagrams.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret basic electrical and electronics concepts.	1.1 <u>Basic electrical terms</u> are defined. 1.2 <u>Common electronic components</u> are identified. 1.3 Safety procedures are followed and hazards are controlled. 1.4 <u>Basic circuit behaviors</u> are explained. 1.5 Measurements are taken using a multimeter and results are recorded.
2. Identify computer components.	2.1 Main computer parts are identified from visual inspection. 2.2 <u>Internal components</u> are named and located. 2.3 <u>External peripherals</u> are recognized. 2.4 <u>Ports and connectors</u> are identified. 2.5 Component functions are described.
3. Identify mobile phone components.	3.1 Basic mobile phone parts are identified through visual inspection. 3.2 <u>Internal component of mobile phones</u> is located and named. 3.3 Connectors, flex cables, and modules are recognized. 3.4 <u>Chip-level components</u> are identified on the board. 3.5 Component functions are described.
4. Interpret mobile phone circuit and schematic diagrams.	4.1 Circuit symbols and legends are identified. 4.2 Signal paths and connections are traced on the diagram. 4.3 Component locations are matched with the schematic. 4.4 <u>Functions of key sections</u> are interpreted. 4.5 Voltage and signal flow are understood from the diagram.

Range of Variables

Variable	Range
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	May include but not limited to:
1. Basic electrical terms	1.1 Voltage 1.2 Current 1.3 Resistance 1.4 Power
2. Common electronic components	2.1 Resistor 2.2 Capacitor 2.3 Diode 2.4 Transistor
3. Basic circuit behaviors	3.1 Series Behavior 3.2 Parallel Behavior 3.3 Active Behavior 3.4 Passive Behavior 3.5 DC Behavior 3.6 AC Behavior
4. Internal components	4.1 Motherboard 4.2 CPU 4.3 RAM 4.4 Storage
5. External peripherals	5.1 Keyboard 5.2 Mouse 5.3 Monitor 5.4 Printer
6. Ports and connectors of mobile phone	6.1 USB 6.2 HDMI 6.3 Audio 6.4 LAN
7. Internal component of mobile phones	7.1 System on a Chip (SoC) 7.2 Motherboard (PCB) 7.3 Battery 7.4 Display Assembly (Screen) 7.5 Memory and Storage (RAM/NAND) 7.6 Camera Modules 7.7 Speakers and Microphone 7.8 Antennas (Cellular, Wi-Fi, GPS)
8. Chip-level components	8.1 System on a Chip (SoC) 8.2 Memory ICs (RAM and Flash Storage) 8.3 Power Management IC (PMIC) 8.4 RF Transceiver IC (Radio Frequency Integrated Circuit) 8.5 Audio Codec IC 8.6 Charging IC 8.7 Display Driver IC
9. Functions of key sections	9.1 Power 9.2 Charging

	9.3 Display
	9.4 Network

Curricular Content Guide

1. Underpinning Knowledge	1.1 Basic electrical and electronic concepts, terms, and simple circuit behavior. 1.2 Identification of common electronic components and their basic functions. 1.3 Understanding of electrical safety rules, hazards, and safe working practices. 1.4 Basic idea of using a multimeter to measure voltage, current, and resistance. 1.5 Identification of main computer components, internal parts, peripherals, ports, and connectors. 1.6 Understanding of the functions of major computer hardware components. 1.7 Identification of mobile phone parts, internal components, connectors, flex cables, and modules. 1.8 Basic understanding of chip-level components on a mobile phone circuit board and their roles.
2. Underpinning Skills	2.1 Interpreting basic electrical and electronics concepts safely. 2.2 Identifying common electronic components and basic circuit behavior. 2.3 Using a multimeter to take and record measurements correctly. 2.4 Identifying main computer components, peripherals, ports, and connectors. 2.5 Identifying mobile phone components at module and chip level. 2.6 Interpreting mobile phone circuit and schematic diagrams. 2.7 Tracing signal flow and understanding functional sections of circuits.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual)

	4.2 Computer/laptop/notebook 4.3 Relevant tools 4.4 Facilities needed to perform the activities 4.5 Required learning materials
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Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment must evidence that the candidate: 1.1 interpreted basic electrical and electronics concepts. 1.2 identified computer components. 1.3 identified mobile phone components. 1.4 interpreted mobile phone circuit and schematic diagrams.
2. Methods of Assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: ASSEMBLE AND CONFIGURE COMPUTER SYSTEMS	Nominal Duration: 40 hrs.	Unit Code: SICIP-ICT-ITM-02-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to assemble and configure computer systems. It specifically includes the tasks of preparing tools, workspace and components, disassembling and assembling computer hardware, configuring BIOS and system settings, installing operating systems, drivers, and software and optimizing system performance and updates.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare tools, workspace, and components.	1.1 Workspace is cleaned and cleared of clutter. 1.2 ESD protection (mat, wrist strap) is set up and verified. 1.3 <u>Required tools</u> are gathered and inspected. 1.4 Tools are tested for functionality and safety. 1.5 Components and parts are checked against the job checklist.
2. Disassemble and assemble computer hardware.	2.1 Power is disconnected and safety is ensured. 2.2 External parts and casing are removed. 2.3 Internal components are detached following proper sequence. 2.4 Parts are stored and handled with ESD precautions. 2.5 Components are reassembled correctly and securely. 2.6 System is checked for proper installation and completeness.
3. Configure BIOS and system settings.	3.1 BIOS interface is accessed using the correct key. 3.2 System information and hardware status are reviewed. 3.3 <u>Boot order settings</u> are adjusted as required. 3.4 Hardware features and security options are configured. 3.5 Changes are saved and system is restarted. 3.6 Configuration results are verified after reboot.
4. Install operating systems, drivers, and software.	4.1 Installation media is prepared and selected. 4.2 <u>Operating system</u> setup is initiated and completed. 4.3 <u>Required drivers</u> are installed and updated. 4.4 <u>Essential software applications</u> are installed. 4.5 System configuration and updates are applied. 4.6 Installation results are checked for proper operation.
5. Optimize system performance and updates.	5.1 System performance issues are identified through basic checks. 5.2 Unnecessary files and programs are removed. 5.3 Startup and background processes are adjusted.

	5.4	System updates and patches are installed.
	5.5	Security and performance settings are optimized.
	5.6	Overall improvements are verified after optimization.

Range of Variables

Variable	Range
	May include but not limited to:
1. Required tools	1.1 Phillips-head Screwdriver 1.2 Flat-head Screwdriver 1.3 Anti-static Wrist Strap 1.4 Needle-nose Pliers 1.5 Tweezers 1.6 Cable Ties/Velcro Straps
2. Boot order settings	2.1 Power on system 2.2 Enter BIOS 2.3 Open Boot Menu 2.4 View boot devices 2.5 Select first boot device 2.6 Arrange boot priority 2.7 Save and exit BIOS 2.8 Verify boot sequence
3. Operating system	3.1 Microsoft Windows 3.2 macOS 3.3 Linux 3.4 Android 3.5 iOS 3.6 Chrome OS
4. Required drivers	4.1 Chipset Drivers 4.2 Graphics Drivers (GPU/Video) 4.3 Audio Drivers (Sound Card) 4.4 Network/Ethernet Drivers 4.5 Wi-Fi/Wireless Drivers 4.6 Storage Drivers (SATA/NVMe/AHCI) 4.7 Bluetooth Drivers 4.8 USB Drivers 4.9 Touchpad/Mouse Drivers 4.10 Printer Drivers
5. Essential software applications	5.1 Web Browser 5.2 Office Suite 5.3 Antivirus Software 5.4 Media Player 5.5 PDF Reader 5.6 Email Client 5.7 File Compression Utility 5.8 Messaging/Communication Software

Curricular Content Guide

1. Underpinning Knowledge	1.1 Concept of preparing a clean, safe workspace and the importance of ESD protection for computer hardware work. 1.2 Identification and purpose of common computer hardware tools and why tool condition matters. 1.3 Basic understanding of computer disassembly and assembly sequence to prevent damage to components. 1.4 Knowledge of safe handling and storage of internal components to avoid static or physical damage. 1.5 Understanding of BIOS functions, system information, boot order, and basic hardware configuration. 1.6 Purpose of device drivers and software applications for proper system operation. 1.7 Basic idea of system performance issues and factors that affect speed and stability. 1.8 Understanding of system updates, security settings, and verification of system optimization results.
2. Underpinning Skills	2.1 Preparing tools, workspace, and components using ESD safety practices. 2.2 Disassembling and assembling computer hardware correctly. 2.3 Handling and storing components safely during maintenance. 2.4 Configuring BIOS and basic system settings. 2.5 Installing operating systems, drivers, and essential software. 2.6 Optimizing system performance and applying updates. 2.7 Verifying system functionality after installation and configuration.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Computer/laptop/notebook 4.3 Multimedia 4.4 Tools (content writing, graphic design, video creation) 4.5 Required learning materials

	4.6 Specifications and work instructions
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Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment must evidence that the candidate: 1.1 prepared tools, workspace, and components. 1.2 disassembled and assemble computer hardware. 1.3 configured BIOS and system settings. 1.4 installed operating systems, drivers, and software. 1.5 optimized system performance and updates.
2. Methods of Assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: MANAGE NETWORK AND INTERNET CONNECTIVITY	Nominal Duration: 40 hrs.	Unit Code: SICIP-ICT-ITM-03-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to manage network and internet connectivity. It specifically includes the tasks of setting up local area network (LAN) and Wi-Fi connections, configuring network devices, monitoring network performance, applying troubleshooting methods for network failures and securing network access and antivirus tools.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Set up local area network (LAN) and Wi-Fi connections.	1.1 LAN requirements are identified. 1.2 Appropriate hardware devices are identified and selected and prepared according to specifications. 1.3 <u>Network cables</u> are installed, connected, and labeled as per standard procedures. 1.4 Wi-Fi router and access points are configured following manufacturer's guidelines. 1.5 <u>Network devices</u> are powered, tested, and verified for operational status. 1.6 Security settings and access restrictions are applied according to organizational policy. 1.7 Connectivity is confirmed and functionality is documented in line with workplace requirements.
2. Configure network devices.	2.1 Configuration requirements are obtained from organization. 2.2 Network devices are accessed using appropriate <u>interfaces</u> and methods. 2.3 Device settings are configured according to specifications and technical guidelines. 2.4 <u>Internet Protocol (IP)</u> addresses, subnet masks, and gateway parameters are assigned as required. 2.5 Routing, switching, and wireless settings are adjusted and saved as per instructions. 2.6 <u>Security features</u> are enabled in line with organizational standards. 2.7 Device configurations are tested and verified for correct operation.
3. Monitor network performance.	3.1 Network monitoring requirements are identified. 3.2 <u>Network monitoring tools</u> are selected and applied according to specifications. 3.3 Traffic patterns and <u>bandwidth</u> utilization are observed

	and recorded as instructed. 3.4 Device logs and performance metrics are reviewed to detect abnormalities. 3.5 Performance data is analyzed and compared against baseline standards. 3.6 Monitoring reports are prepared and submitted according to workplace requirements.
4. Apply troubleshooting methods for network failures.	4.1 Network failure symptoms are identified from monitoring data and user reports. 4.2 Hardware and software components are checked to locate the source of the problem. 4.3 Fault isolation methods are applied according to standard troubleshooting procedures. 4.4 Configuration settings are verified and corrected as required. 4.5 Diagnostic tools are used to test connectivity and network performance. 4.6 Solutions are implemented and tested to confirm successful restoration of services.
5. Secure network access and antivirus tools.	5.1 Organizational security requirements are identified. 5.2 User access rights and permissions are configured according to authorization levels. 5.3 Network authentication methods are applied as instructed. 5.4 Firewalls and security rules are enabled to protect against unauthorized access. 5.5 Antivirus software is installed and updated according to vendor specifications. 5.6 Detected threats and malicious activities are quarantined or removed using approved procedures.

Range of Variables

Variable	Range
	May include but not limited to:
1. Network cables	1.1 UTP 1.2 STP 1.3 Fiber optic cables 1.4 Patch chords
2. Network devices	2.1 Switches 2.2 Hubs 2.3 Routers 2.4 Network interface cards (NICs) 2.5 Access Point (AP) 2.6 Firewall Device 2.7 Modem 2.8 Patch panels
3. Interfaces	3.1 Command line interface (CLI)

	3.2 Graphical user interface (GUI) 3.3 Web interface 3.4 Console cable
4. Internet Protocol (IP)	4.1 TCP/IP 4.2 DHCP 4.3 DNS 4.4 VLAN 4.5 NAT 4.6 routing protocols (RIP, OSPF, EIGRP).
5. Security features	5.1 Access control lists (ACLs) 5.2 Firewall rules 5.3 Port security 5.4 Encryption (WPA2/WPA3) 5.5 Strong password policies
6. Network monitoring tools	6.1 SNMP tools 6.2 Wireshark 6.3 PRTG 6.4 Nagios 6.5 SolarWinds 6.6 MRTG 6.7 Built-in device monitoring utilities
7. Bandwidth	7.1 Bandwidth usage 7.2 Latency 7.3 Jitter 7.4 Packet loss 7.5 Throughput 7.6 Uptime/downtime
8. Device logs	8.1 System logs 8.2 Event logs 8.3 Performance counters 8.4 Real-time dashboards
9. Software components	9.1 Firmware 9.2 Drivers 9.3 Operating system network settings 9.4 Firewall software
10. Fault isolation methods	10.1 Fault isolation 10.2 Step-by-step testing 10.3 Divide-and-conquer 10.4 Substitution method 10.5 Escalation procedures
11. Diagnostic tools	11.1 Ping 11.2 Traceroute 11.3 Ipconfig/ifconfig 11.4 Netstat 11.5 Cable tester 11.6 Wireshark 11.7 SNMP monitors

12. Antivirus software	12.1 Windows Defender 12.2 Kaspersky 12.3 McAfee 12.4 Avast 12.5 Norton 12.6 Enterprise endpoint security solutions
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Curricular Content Guide

1. Underpinning Knowledge	1.1 Basic concept of Local Area Network (LAN) and Wi-Fi networks. 1.2 Understanding of common network hardware devices and their roles in wired and wireless networking. 1.3 Knowledge of standard network cabling, connections, labeling practices, and their purpose. 1.4 Basic idea of configuring network devices, Internet Protocol (IP) addressing, subnet masks, and gateways. 1.5 Understanding of routing, switching, and wireless settings required for network connectivity. 1.6 Concept of network security settings, access control, and organizational security policies. 1.7 Basic understanding of network monitoring, traffic observation, bandwidth usage, and performance indicators. 1.8 Knowledge of common network failure symptoms, fault isolation concepts, and troubleshooting approaches.
2. Underpinning Skills	2.1 Setting up LAN and Wi-Fi networks according to specifications. 2.2 Installing, connecting, and testing network hardware and cables. 2.3 Configuring network devices and IP addressing parameters. 2.4 Applying network security settings and access controls. 2.5 Monitoring network performance using appropriate tools. 2.6 Troubleshooting network faults and restoring connectivity. 2.7 Implementing antivirus protection and securing network access.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided:

	4.1 Workplace (simulated or actual) 4.2 Computer/laptop 4.3 Specifications and work instructions 4.4 Learning materials 4.5 Relevant equipment, software and facilities needed to perform the activities
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Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment must evidence that the candidate: 1.1 set up local area network (LAN) and Wi-Fi connections. 1.2 configured network devices. 1.3 monitored network performance. 1.4 applied troubleshooting methods for network failures. 1.5 secured network access and antivirus tools.
2. Methods of Assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: MAINTAIN DATA BACKUP AND SYSTEM SECURITY	Nominal Duration: 60 hrs.	Unit Code: SICIP-ICT-ITM-04-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to maintain data backup and system security. It specifically includes the tasks of Identifying critical data and establish backup schedules, performing data backup using appropriate storage methods (cloud or local), testing backup integrity to ensure data, recovering lost or corrupted data and implementing disaster recovery procedures.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify critical data and establish backup schedules.	1.1 Organizational requirements for data backup are identified. 1.2 <u>Backup methods</u> are selected according to organizational and technical requirements. 1.3 Backup schedules are developed as per workplace standards. 1.4 <u>Backup tools</u> and <u>storage media</u> are identified and prepared as specified.
2. Perform data backup using appropriate storage methods (cloud or local).	2.1 Backup requirements are identified. 2.2 Appropriate storage methods (cloud or local) are selected according to data type and security needs. 2.3 Backup tools are configured as specified. 2.4 <u>Data</u> is backed up using approved methods and verified for accuracy and completeness. 2.5 Backup files are secured and stored in line with organizational standards.
3. Test backup integrity to ensure data.	3.1 Organizational requirements for backup testing are identified. 3.2 Backup media and storage devices are accessed using approved methods. 3.3 Test restorations are performed to verify the accuracy and completeness of backup data. 3.4 Backup files are validated using <u>verification tools</u> and checksum <u>methods</u> . 3.5 Restored data is compared with original data to ensure integrity and consistency.
4. Recover lost or corrupted data.	4.1 Data recovery requirements are identified. 4.2 <u>Recovery tools</u> and <u>methods</u> are selected according to the type and extent of data loss. 4.3 Recovery processes are applied to retrieve lost or corrupted files.

	4.4 Data is restored to designated systems. 4.5 Restored data is verified for accuracy, completeness, and usability. 4.6 <u>Security measures</u> are applied during and after recovery to protect sensitive information.
5. Implement disaster recovery procedures	5.1 Disaster recovery requirements are identified. 5.2 <u>Disaster recovery plans</u> are reviewed and updated. 5.3 Recovery roles and responsibilities are assigned and communicated. 5.4 <u>Recovery procedures</u> are executed using approved methods and tools. 5.5 Critical systems and data are restored to operational status within required timeframes. 5.6 System functionality is tested and verified after recovery activities.

Range of Variables

Variable	Range May include but not limited to:
1. Backup methods	1.1 Full backup 1.2 Incremental backup 1.3 Differential backup 1.4 Cloud backup 1.5 Snapshot backup
2. Backup tools	2.1 Windows Backup 2.2 Veeam 2.3 Acronis 2.4 Commvault 2.5 Google Drive/OneDrive 2.6 Enterprise backup systems
3. Storage media	3.1 Local disks 3.2 External hard drives 3.3 NAS devices 3.4 Cloud storage 3.5 Tape drives
4. Data	4.1 Databases 4.2 Application files 4.3 Emails 4.4 Multimedia 4.5 Business-critical records
5. Verification tools	5.1 Backup software validation features 5.2 Third-party integrity checkers 5.3 Command-line utilities
6. Methods	6.1 Checksum/MD5/SHA validation 6.2 File comparison 6.3 Log file review

	6.4 Test restore
7. Recovery tools	7.1 Recuva 7.2 EaseUS 7.3 Stellar Data Recovery 7.4 Ontrack 7.5 TestDisk 7.6 Built-in OS recovery utilities
8. Recovery methods	8.1 Data recovery software 8.2 Backup restoration 8.3 File system repair 8.4 RAID recovery 8.5 Cloud recovery tools 8.6 Professional recovery services
9. Security measures	9.1 Encryption 9.2 Access control 9.3 Antivirus scanning after recovery 9.4 Secure transfer protocols.
10. Disaster recovery plans	10.1 Business continuity plan (BCP) 10.2 Disaster recovery plan (DRP) 10.3 Emergency response plan
11. Recovery procedures	11.1 Backup restoration 11.2 System rebuilds 11.3 Cloud-based recovery 11.4 Failover procedures 11.5 Redundant systems utilization

Curricular Content Guide

1. Underpinning Knowledge	1.1 Concept of critical data and why regular data backup is necessary for organizations. 1.2 Understanding of different backup methods and backup schedules used to protect data. 1.3 Knowledge of backup tools, storage media, and the difference between cloud and local storage. 1.4 Basic idea of performing data backups and verifying backup accuracy and completeness. 1.5 Understanding of securing backup files to prevent unauthorized access or data loss. 1.6 Concept of backup testing and why test restoration is required to ensure data integrity. 1.7 Knowledge of data recovery principles for lost, deleted, or corrupted files. 1.8 Understanding of security measures required during data recovery activities.
2. Underpinning Skills	2.1 Identifying critical data and defining backup requirements. 2.2 Selecting backup methods, tools, and storage media.

	2.3 Creating and following data backup schedules. 2.4 Performing data backups using cloud or local storage. 2.5 Testing backup integrity through verification and restoration. 2.6 Recovering lost or corrupted data safely and accurately. 2.7 Implementing disaster recovery procedures and verifying system restoration.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace 3.8 Good relationships with peers, sub-ordinates and seniors in workplace
4. Resource Implications	Following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Computer/laptop 4.3 Job specifications and work instructions 4.4 Relevant tools, equipment and facilities 4.5 Required learning materials

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment must evidence that the candidate: 1.1 identified critical data and establish backup schedules. 1.2 performed data backup using appropriate storage methods (cloud or local). 1.3 tested backup integrity to ensure data. 1.4 recovered lost or corrupted data. 1.5 implemented disaster recovery procedures.
2. Methods of Assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PRACTICE SOLDERING AND DE-SOLDERING	Nominal Duration: 40 hrs.	Unit Code: SICIP-ICT-ITM-05-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to practice soldering and re-soldering. It specifically includes the tasks of selecting soldering tools and materials, performing soldering on PCB, performing de-soldering components from PCB, performing re-soldering from PCB and cleaning soldering tools and workspace.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Select soldering tools and materials	1.1 <u>Soldering tools</u> are selected based on the specific repair task requirements. 1.2 <u>Soldering materials</u> are selected according to the type of components being soldered. 1.3 Soldering tools are used according to safety guidelines to prevent damage.
2. Perform soldering on PCB	2.1 Safety precautions are followed during soldering to avoid component damage. 2.2 PCB components are positioned and secured in place to ensure correct alignment before soldering. 2.3 Soldering iron is heated to the appropriate temperature, 2.4 Soldering is carried out on the PCB pads and component leads. 2.5 Solder joints are inspected for quality. 2.6 Excess flux is cleaned from the PCB after soldering to prevent corrosion or damage.
3. Perform de-soldering components from PCB	3.1 Safety measures are followed, ensuring adequate ventilation. 3.2 Component leads are heated with the soldering iron to melt the solder and prepare for removal. 3.3 Soldering hot air gun is applied to remove the molten solder and safely detach the component. 3.4 Component is carefully removed without causing damage to the PCB. 3.5 Excess solder and residual flux are cleaned from the PCB after component removal.
4. Perform re-soldering from PCB	4.1 PCB is inspected for faulty solder joints or components that require re-soldering. 4.2 Faulty component or joint is heated. 4.3 Old solder is removed using solder wick. 4.4 New solder is applied to the PCB.

	4.5 Re-soldered joint is inspected for strength, proper connection, and any signs of cold solder joints or shorts.
	4.6 PCB is cleaned of any flux residue after re-soldering.
5. Clean soldering tools and workspace	5.1 Soldering tools are cleaned using appropriate cleaning materials.
	5.2 Soldering iron tips are wiped clean and stored properly to avoid oxidation.
	5.3 Workspace is cleared of any excess materials.

Range of Variables

Variable	Range May include but not limited to:
1. Soldering tools	1.1 Soldering stand 1.2 Soldering iron 1.3 PCB Holder 1.4 Tweezers 1.5 Point cutter 1.6 Nose pliers 1.7 ESD mat
2. Soldering materials	2.1 Soldering paste 2.2 Soldering lead 2.3 IC paste 2.4 Glue 2.5 Jumper wire 2.6 Heat tape 2.7 Both tape

Curricular Content Guide

1. Underpinning Knowledge	1.1 Basic concept of soldering and its purpose in joining electronic components to a PCB. 1.2 Identification of common soldering tools and materials and their appropriate uses. 1.3 Understanding of soldering safety principles, heat control, and protection of PCB components. 1.4 Concept of good and poor solder joints and common soldering defects. 1.5 Understanding of de-soldering principles and safe removal of components from PCBs. 1.6 Knowledge of re-soldering concepts to repair faulty joints or replace components. 1.7 Purpose of flux use and the importance of cleaning flux residues from PCBs.
2. Underpinning Skills	2.1 Selecting appropriate soldering tools and materials. 2.2 Applying safety practices during soldering and de-soldering.

	2.3 Performing soldering on PCB components correctly. 2.4 Removing components using de-soldering techniques safely. 2.5 Re-soldering PCB joints and components accurately. 2.6 Inspecting solder joints for quality and defects. 2.7 Cleaning soldering tools, PCBs, and the workspace after work.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Computer/laptop 4.3 Learning materials 4.4 Specifications and work instructions 4.5 Relevant tools, equipment and facilities

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment must evidence that the candidate: 1.1 selected soldering tools and materials 1.2 performed soldering on PCB 1.3 performed de-soldering components from PCB 1.4 performed re-soldering from PCB 1.5 cleaned soldering tools and workspace
2. Methods of Assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: TEST COMPONENTS AND SERVICE MOBILE PHONE	Nominal Duration: 70 hrs.	Unit Code: SICIP-ICT-ITM-06-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to test components and service mobile phone. It specifically includes the tasks of preparing mobile phone tools, and workspace for servicing, applying standard testing methods for components, disassemble the mobile phone, testing components in-circuit and off-circuit, replacing faulty or damaged components and re-assembling the mobile phone.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare mobile phone, tools, and workspace for servicing.	1.1 Mobile phone is inspected thoroughly to assess the type of servicing required. 1.2 <u>All necessary tools and equipment</u> are gathered and checked for proper functionality. 1.3 Service area is prepared by ensuring it is clean, and organized. 1.4 Repair service manual is identified to ensure all necessary steps are followed and completed.
2. Apply standard testing methods for components.	2.1 <u>Standard testing criteria</u> for each electronic component are understood. 2.2 Each component's specifications are understood to ensure proper testing. 2.3 <u>Testing tools</u> is identified. 2.4 <u>Appropriate testing methods</u> for each component are recognized. 2.5 Testing method is chosen to match the type of component. 2.6 Testing methods are followed to ensure that the results align with component specifications.
3. Disassemble the mobile phone.	3.1 <u>Mobile phone</u> is carefully examined to identify the components that need to be disassembled. 3.2 Mobile phone is powered off. 3.3 Screws, clips, and adhesives securing the phone's outer casing are identified and removed. 3.4 Back cover, screen, or other components are gently separated. 3.5 All internal components are carefully disconnected. 3.6 Disassembled parts are organized and placed in a safe, clean area to avoid loss or damage.

4. Test components in-circuit and off-circuit.	4.1 Testing method is selected based on the type of component being tested. 4.2 In-circuit testing is performed. 4.3 Off-circuit testing is performed. 4.4 Test results are interpreted to determine if the component functions correctly. 4.5 Corrective actions are carried out as needed after testing.
5. Replace faulty or damaged components.	5.1 Faulty components are identified through testing or inspection. 5.2 Necessary replacement parts are selected. 5.3 Faulty component is carefully removed without damaging surrounding parts. 5.4 Replacement component is checked for compatibility with the mobile phone. 5.5 New component is installed and securely connected to the relevant parts.
6. Reassemble the mobile phone.	6.1 Mobile phone components are carefully reassembled following the manufacturer's guidelines. 6.2 Internal components are reconnected securely, ensuring proper alignment and connection. 6.3 Screen, back cover, or any external parts are carefully positioned and reattached. 6.4 All screws, clips, and fasteners are tightened or secured to ensure the phone is fully assembled. 6.5 Mobile phone is powered on to verify that all components are functioning correctly after reassembly. 6.6 The phone's buttons, screen, ports, and other external features are tested.

Range of Variables

Variable	Range May include but not limited to:
1. All necessary tools and equipment	1.1 Screw driver 1.2 Tweezer 1.3 Casing opener 1.4 PCB holder 1.5 Opener 1.6 Nose pliers 1.7 Combination pliers 1.8 Cutting pliers 1.9 Blade cutter 1.10 Point cutter 1.11 Soldering stand 1.12 Soldering iron

	1.13 PCB Holder 1.14 Tweezers
2. Standard testing criteria	2.1 Verify component values (resistance, capacitance, inductance) within tolerance. 2.2 Ensure voltage, current, and power ratings are within limits. 2.3 Test for leakage currents and thermal stability. 2.4 Check switching speed and reliability under stress. 2.5 Perform functional and lifespan testing for durability. 2.6 Ensure compliance with safety and environmental standards.
3. Testing tools	3.1 Avo meter/multimeter 3.1.1 Analog 3.1.2 Digital 3.2 DC Power Supply 3.3 GSM/3G/4G/5G Signal Tester 3.4 LCD Tester
4. Appropriate testing methods	4.1 Resistance test 4.2 Continuity test 4.3 Voltage test
5. Mobile phone	5.1 Basic Phone (Feature phone) 5.2 Smartphone/ Android Phone/ iPhone (iOS Phone)

Curricular Content Guide

1. Underpinning Knowledge	1.1 Basic concept of preparing a mobile phone, tools, and a clean workspace before servicing. 1.2 Understanding of how visual inspection helps identify the type of mobile phone fault or service need. 1.3 Knowledge of standard testing concepts used for electronic components in mobile phones. 1.4 Understanding of component specifications and why correct testing methods matter. 1.5 Concept of in-circuit and off-circuit testing and how test results indicate component condition. 1.6 Basic understanding of safe mobile phone disassembly and identification of internal components. 1.7 Knowledge of faulty component identification and compatibility requirements for replacement parts. 1.8 Concept of proper reassembly following manufacturer guidelines to ensure correct alignment and connection.
2. Underpinning Skills	2.1 Preparing the mobile phone, tools, and workspace for servicing. 2.2 Applying standard testing methods for electronic components. 2.3 Disassembling mobile phones safely and systematically. 2.4 Testing components in-circuit and off-circuit.

	2.5 Identifying and replacing faulty or damaged components. 2.6 Reassembling the mobile phone according to manufacturer guidelines. 2.7 Verifying functionality of the phone after servicing and repair.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Computer/laptop 4.3 Relevant tools, equipment and facilities 4.4 Required learning materials

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment must evidence that the candidate: 1.1 prepared mobile phone, tools, and workspace for servicing. 1.2 applied standard testing methods for components. 1.3 disassembled the mobile phone. 1.4 tested components in-circuit and off-circuit. 1.5 replaced faulty or damaged components. 1.6 reassembled the mobile phone.
2. Methods of Assessment	Methods of assessment may include but is not limited to: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: APPLY SOFTWARE INSTALLATION AND FLASHING MOBILE PHONE	Nominal Duration: 40 Hrs.	Unit Code: SICIP-ICT-ITM-07-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply software installation and flashing mobile phone. It specifically includes the tasks of identifying software requirements of mobile phone, preparing flashing tools and install drivers, connecting mobile phone to PC or flashing tool, flashing mobile phone software and cleaning tools and equipment.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify software requirements of mobile phone	1.1 Software requirements for installing and flashing the mobile phone are identified. 1.2 <u>Appropriate flashing tools</u> and <u>software device</u> are identified based on the mobile phone brand and model. 1.3 Mobile phone's model and specifications are checked. 1.4 Required version of the mobile phone's operating system or firmware is identified and verified. 1.5 System requirements for flashing are checked and confirmed.
2. Prepare flashing tools and install drivers	2.1 Necessary flashing tools are gathered based on the mobile phone model and software requirements. 2.2 Flashing tool is installed on the computer. 2.3 Correct version of flashing software is selected. 2.4 Drivers required for the flashing process are identified and downloaded from reliable sources. 2.5 Drivers are installed on the computer. 2.6 Installation of flashing tools and drivers is confirmed and checked.
3. Connect mobile phone to PC or flashing tool	3.1 Mobile phone is powered off before connecting it to the PC or flashing tool to avoid any electrical damage. 3.2 Appropriate USB cable is selected. 3.3 Mobile phone is connected to the PC using the USB cable. 3.4 Flashing tool or software is launched on the PC, and the mobile phone is detected in the tool's interface. 3.5 Any necessary drivers are installed on the PC. 3.6 PC or flashing tool is configured to detect the connected mobile phone, and the connection is tested for stability.
4. Flash mobile phone software	4.1 Mobile phone is prepared by ensuring it is fully charged, in the correct mode. 4.2 Flashing tool is configured to the appropriate settings,

	including the correct partition and flashing method. 4.3 Mobile phone is detected by the flashing software, and any necessary drivers are verified. 4.4 Mobile phone flashing is carried out as requirement. 4.5 Once the flashing process is completed, the mobile phone is automatically rebooted. 4.6 Mobile phone is tested to confirm that the new software is functioning correctly.
5. Clean tools and equipment	5.1 Tools and equipment are inspected to ensure they are free of dirt, debris, or residue. 5.2 Appropriate cleaning materials are selected based on the type of tool or equipment. 5.3 Tools and equipment are carefully cleaned to remove any dirt, grease, or contaminants. 5.4 Work area is cleaned to prevent tools or equipment from becoming contaminated again. 5.5 Cleaned tools and equipment are stored.

Range of Variables

Variable	Range (May includes but not limited to):
1. Appropriate flashing tools	1.1 Odin flash tool 1.2 SP flash tool 1.3 SPD flash tool 1.4 Qualcomm flash tool 1.5 MTK flash tool
2. Software device	2.1 Octoplus 2.2 Z3X 2.3 UMT 2.4 Miracle 2.5 NCK Dongle

Curricular Content Guide

1. Underpinning Knowledge	1.1 Basic concept of mobile phone software, firmware, and operating system used in different phone models. 1.2 Understanding of software requirements needed for mobile phone flashing and installation. 1.3 Knowledge of mobile phone model identification and how specifications affect software selection. 1.4 Concept of flashing tools, flashing software, and their compatibility with different mobile phone brands. 1.5 Understanding of system and driver requirements for successful flashing operations. 1.6 Concept of flashing modes, partitions, and correct flashing settings.
2. Underpinning Skills	2.1 Identifying mobile phone software and firmware requirements.

	2.2 Selecting appropriate flashing tools and software for specific models. 2.3 Installing and verifying flashing tools and required drivers. 2.4 Connecting mobile phones to PCs or flashing devices safely. 2.5 Configuring and performing mobile phone software flashing. 2.6 Verifying software installation and post-flash functionality. 2.7 Cleaning, maintaining, and storing tools and equipment properly.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Relevant tools, Equipment, software and facilities needed to perform the activities. 4.3 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified software requirements of mobile phone. 1.2 prepared flashing tools and install drivers. 1.3 connected mobile phone to PC or flashing tool. 1.4 flashed mobile phone software. 1.5 cleaned tools and equipment.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

END OF COMPETENCY STANDARD

Workshop/Lab Facility Standard

Course Name:	IT Support Technique & Mobile Phone Servicing
Number of Trainees:	25

Course-wise Training Space (Theoretical Classroom, Workshop/ Lab/ Classroom cum Workshop):

- Classroom – 350 sft (33 sqm)
- Workshop/ lab – 800 sft (75 sqm)
- OR
- Classroom cum workshop – 1000 sft (93 sqm)

Major Training Equipment and Training Facilities:

Sl. No.	Major Equipment and Training facilities	Required facilities
1.	Computer/Laptop	01
2.	Printer	01
3.	Scanner	01
4.	Computers/Laptop: (Minimum Intel Core i5 processor (8 th Gen), 8GB RAM (16GB preferred), 500GB SSD/HDD storage (1TB preferred), High-resolution monitors (19/20-inch minimum, with color accuracy).	25
5.	Projector: High-resolution projector (HD or 4K) for instructor presentations, large screen for the room to ensure visibility for all trainees.	01
6.	Network Devices (Switches and Wi-Fi Routers)	05 sets
7.	Network tools (Clamping tools and cable testers)	10 sets each
8.	Network Cables (UTP and Fiber Optics)	100 meters
9.	Connectors (RJ 45, MTP/MPO)	100
10.	Ergonomic chairs and desks	25
11.	Internet connectivity with power back-up	01
12.	Soldering Stations	25
13.	Soldering Irons	25
14.	Hoyt air gun	25
15.	Multimeters	25
16.	Power Supply Units	25
17.	Battery Testers	05
18.	Screen Testing Jigs (for LCD and touchscreens)	01
19.	USB Testers (for current and voltage checks)	05
20.	Screwdrivers Set (Precision set, including Phillips and flathead)	25
21.	Tweezers (ESD-safe, various sizes)	25

22.	Opening Tools	25
23.	Suction Cups (for screen removal)	25
24.	Magnifying Glasses with Lamp	25
25.	Defective Mobile Phone	15
26.	New Mobile Phone	10

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on IT Support Technique & Mobile Phone Servicing, the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.